# Observation of instability contagion in foreign exchange markets using PUCK analysis

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**Abstract**

We introduce a new formulation of time series analysis based on the PUCK model, which enables quantitative characterization of stability of markets as a function of time. We analyze a set of foreign exchange market's time series and show how stable markets became unstable when big events occurred.

**Keyword:** Econophysics, foreign exchange markets, PUCK model

## 1. Introduction

Market prices are often approximated by simple random walks, however, it has been proved by high frequency data analysis that they are better approximated by random walks with a non-trivial time-dependent market potential [1]. Theoretically this potential force is known to be caused by the herding behavior of trend-follow of dealers [2], and the time series analysis of observing the potential is called the PUCK analysis, which can be used as a real-time tool specifying market stability.

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## 2. Method

We apply a kind of moving average transformation to the given time series, then we estimate the market potential coefficient from the scatter plots of market price changes versus deviations of market's mid-prices from the moving averages. We observe time evolution of the market potential coefficients for several markets in the foreign exchanges using the data provided by EBS, the largest foreign exchange markets provider. Focusing on some events such as the intervention or the big earthquake, we can quantitatively describe development of instability of markets.

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